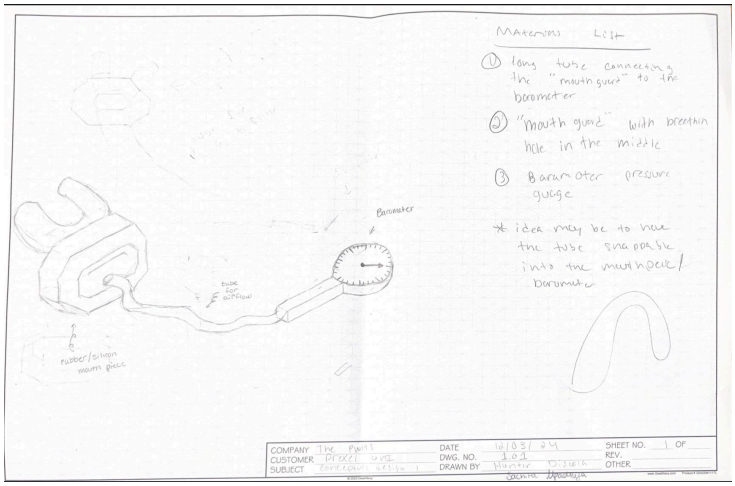
Morphological Chart:

Function	Concept 1	Concept 2	Concept 3
Get the Data (system)	Pressure Gauge	Arduino Microcontroller	Water Mechanical
Measure Data	Gauge (Locked Dial)	Water measure	LCD monitor
Log Data	Write manually	Display on an lcd	Store the data and display on an lcd
Transportation	Case that has a mold for it	Carry with our hands	Plastic bag
Body	Rectangular prism	Just everything out in the open	Pyramid

We need to create a reusable, durable, accurate, and affordable MIP device for the Drexel cardiovascular and pulmonary physical therapy class. This device would help give the students necessary experience to help patients in the process of measuring their maximal inspiratory pressure and strengthening their lungs with an IMT device. The device must be able to be produced on a tight budget and be replicable with every device being precise and accurate, we must be able to transport the device(s) to Drexel without issue.

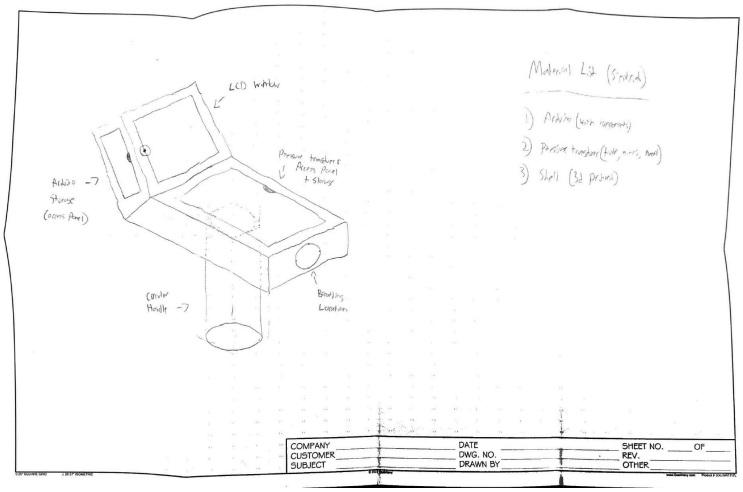
Preliminary Design Concepts

Design A - Mouthpiece Millionaire



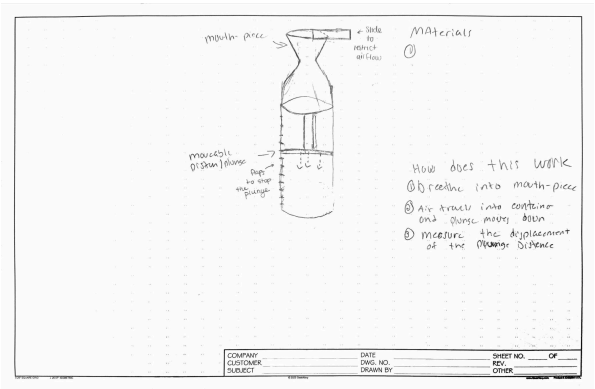
This device will use a football mouthpiece attached to a tube that is also attached to a barometer to measure the air pressure of the person who is using the device. The barometer will display for a split second the MIP of the user. The mouthpieces will be 3D printed out of a durable plastic to be mass producible and able to stand the test of time. Potential drawbacks include the need for washing/cleaning in between usages. The design is easy to use, make, and strong, but it is unreliable and unprofessional looking.

Design B - Arduino Arbol



This design uses an Arduino set up to read the data output by a person breathing into a tube attached to a pressure sensor and display the MIP of the user on a lcd screen. It will be battery powered and able to be transported easily, all components will be housed within a body made of 3D printed plastic. It could increase the cost and time to make the device to add Arduino support and create a custom 3D printed body. The design is hypothetically going to be quite precise due to the fact that it uses electronics instead of analog components, it is also going to be easier to read the data since the device will store the highest data point recorded and display it.

Design C - Air Column Cheetah



This last design is a mechanical air-tube design. A user blows into the tube and a piston is pushed down which compresses the air in the shaft. There are tick marks on the shaft which are calibrated to cm/H2O. A funnel mouthpiece will be used to allow an efficient connection to the mouth to allow maximum airflow.

		Mouthpiece Millionaire	Arduino Arbol	Air Column Cheetah

Criteria	Weight	Score	Weighted Score	Score	Weighted Score	Score	Weighted Score
System (Output Device)	15%	4	0.60	5	0.75	3	0.45
Transportation	5%	5	0.25	5	0.25	5	0.25
Durability	15%	4	0.60	3	0.45	3	0.45
Affordability	15%	4	0.60	3	0.45	5	0.75
Accuracy	25%	4	1.00	5	1.25	3	0.75
Replicability	12.5%	4	0.50	3	0.38	2	0.25
Aesthetics	2.5%	3	0.08	4	0.10	1	0.03
Cleaning	10%	4	0.40	4	0.40	4	0.40
	100%		4.03		4.03		3.33

Conceptual Design Rationale

The process of choosing our main concept had many factors. One being the decision matrix. Our group came up with 8 criteria we found most prominent, and made a percentage of most important. We came up with accuracy being the most important. Our group then scored each design through each criteria on a score 1-4. Coming out on top, was the Mouthpiece Millionaire, and Arduino Arbol, both tied with a weighted score of 4.03. Another factor to our decision processes was the input of Dr. Sarver.

Based on our designs and materials accessible to us we have chosen **Mouthpiece Millionaire (MM)** . Even though the decision matrix had them as the same exact score. The design chosen has the lowest cost to make both in terms of price and time. The MM is also very simple to make, with just 3 simple pieces. One, we have the 3D printed mouthpiece, similar to a snorkel mouthpiece. Then, the mouth-piece will snap into a 3-D printed tube, where the air flows

into a pressure gauge. Although the mouth-piece and tube are both very economically friendly, the barometer runs for aprx. $15-$30. This would total us to aprx. $40-50 for the MM. Our goal is to create the most accurate, cheapest, and durability device, and the MM does all of that well.

After getting feedback from Dr. Sarver, we decided that using an arduino to make a MIP would be very complicated. It would be difficult to wire and keep compact inside of the device. Additionally, cleaning the arduino would not be easy, due to the possibility of having the cables get wet. Calibrating the pressure sensor with another device that we know the pressure reading of would be difficult. This would be unnecessary with the chosen design, which utilizes a pre-calibrated gauge. The arduino is also about $45. We want replicability and an economically sound device, this is neither of them. The difficulty of duplicating these would be extravagant, due to the extensive code and wiring system.

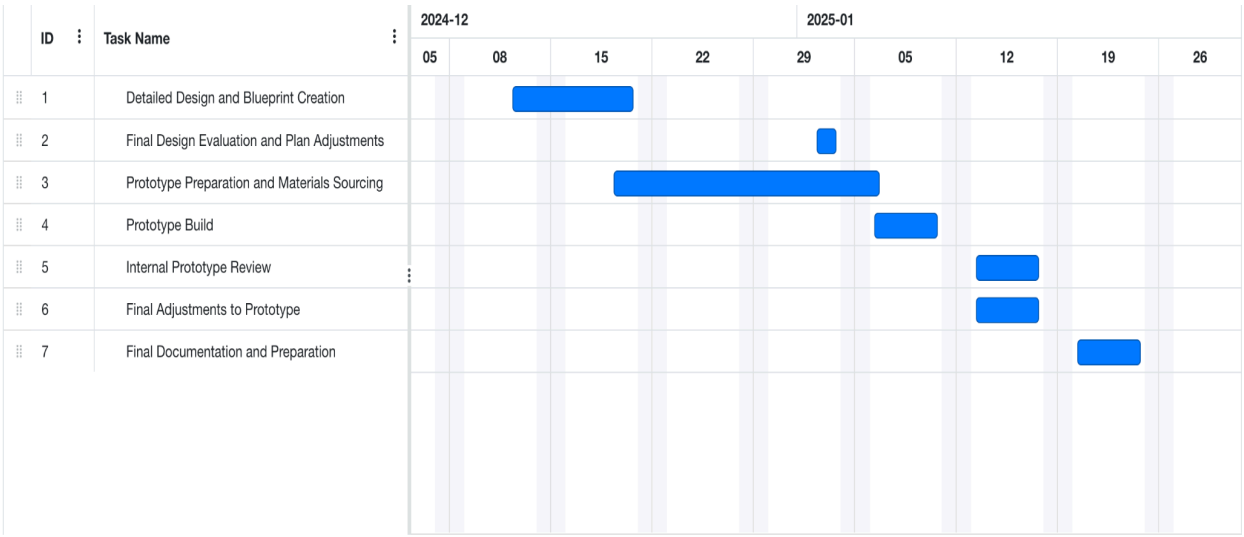
Design C would have been very hard to make and calibrate and got a very low score based on our criteria and the assigned weights. The CC would not be an accurate design. The CC would have notches which the piston latches onto each level as air is blown onto it. The notches would restrict the piston from being pushed further down, affecting the measurements taken. The CC ranked last in our decision matrix, with a weighted score of 3.33. Although the CC would be extremely cheap, it is not accurate nor ideal.

The CAD of the final design is included at the bottom.

PLAN OF ACTION

To start, we should look at the available mouthpieces to choose the best for our intended use that fits within our price range. A 3d printed mouthpiece should work the best, being mass producible and reliable. Also, it could be expensive to buy multiple . After the mouthpiece is acquired we should find a tube that will work and connect to the mouthpiece with minimal cutting. We should then attach the tubing to the mouthpiece and find a way to attach a barometer to the tubing. We should find something with a reliable MIP to calibrate the device and account for any loss of pressure and poor sealing. After the device has been properly calibrated we should look at the design and make adjustments.

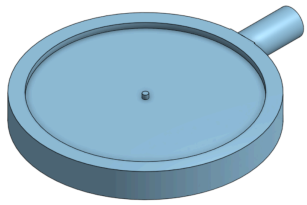
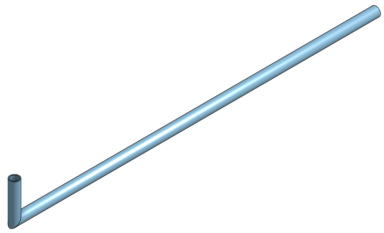
Gantt Chart

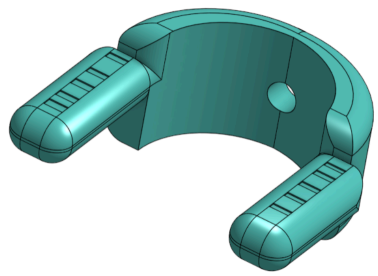


We anticipate making our sketch into an actual MIP that measures the inspiratory pressure of the user will be difficult as we would need to incorporate a barometer into the

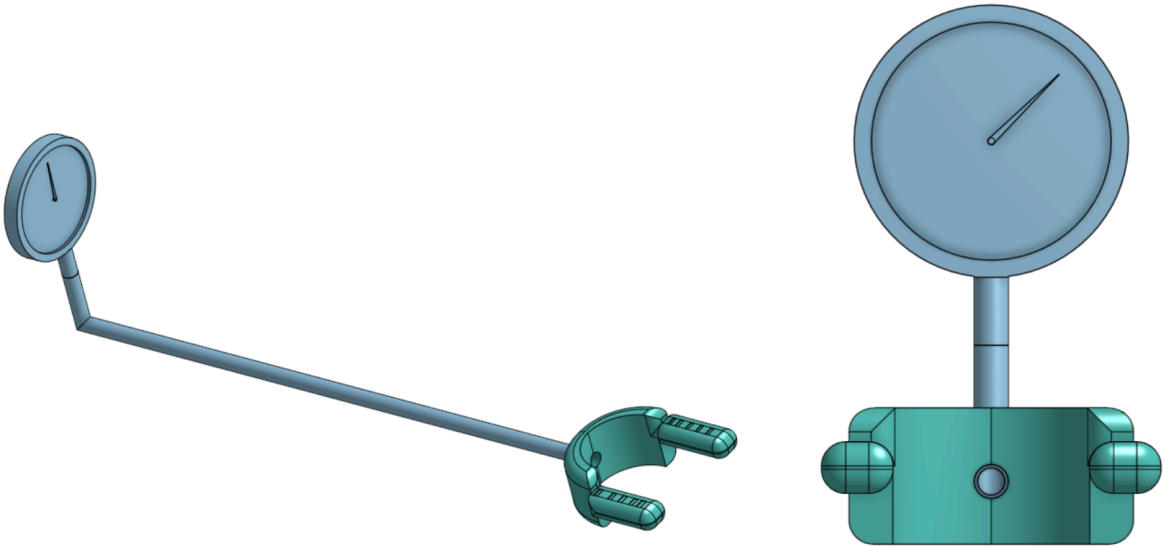
inhalation tube and create a cumulative total of the air pressure inhaled. We will make sure to troubleshoot with the utmost precision and test it rigorously to make sure it works.

Onshape Design - Different Parts and Write-Up

Onshape Part - Name	Onshape Part - Picture	Write-Up
Barometer Dial (without dial)		The picture portrays the barometer of the device. This barometer is connected to the tube (depicted below) and shows the air pressure of the breath.
Tube		This picture depicts the tube that the air will pass through so that the barometer can calculate the pressure. It will be made of stiff material and acts as a pipe out of someone's mouth.

Barometer Dial		This is the dial that will be fixed in the barometer. It will be used to find the air pressure in our design.
Mouthpiece		This picture depicts the mouthpiece. Inspired by a snorkel mouthpiece, this will be put into the user's mouth to allow them to use the device. With this part in their mouth, they will be able to blow into the tube, which will then transfer to the barometer to measure the pressure.

Assembly



This depicts the assembled prototype. As shown here, the mouthpiece, which is attached to the tube, will be put in the mouth of the user. The air will travel through the tube to reach the dial which will show the reading. The dial is connected to the tube with the threaded screw. Everything except the dial is 3D printed.